

Yash Vekaria

Research Interests

Web and **AI** privacy, Security, Transparency, Technology Policy, Internet Measurements

Education

- 2021–Present **Ph.D. in Computer Science** 3.98/4.00
University of California, Davis
- 2021–2023 **MS in Computer Science** 4.00/4.00
University of California, Davis
- 2016–20 **B.Tech. in Computer Science** 9.04/10.00
The LNM Institute of Information Technology (LNMIIT), India.

Publications

- 2026 **SoK: Advances and Open Problems in Web Tracking**
Yash Vekaria, Yohan Beugin, Shaoor Munir, Gunes Acar, Nataliia Bielova, Steven Englehardt, Umar Iqbal, Alexandros Kapravelos, Pierre Laperdrix, Nick Nikiforakis, Jason Polakis, Franziska Roesner, Zubair Shafiq, Sebastian Zimmeck
Under Submission, 2026
- 2026 **FP-Agent: Fingerprinting AI Browsing Agents**
Ethan Wang, Zubair Shafiq, Yash Vekaria* (*primary advisor role)
Under Submission, 2026
- 2026 **Tracking Conversations: Measuring Content and Identity Exposure on AI Chatbots**
Muhammad Jazlan, Ethan Wang, Yash Vekaria, Zubair Shafiq
Under Submission, 2026
- 2026 **SST-Guard: Detecting and Characterizing Server-Side Google Analytics in the Wild**
Muhammad Jazlan, Alexander Gamero-Garrido, Zubair Shafiq, Yash Vekaria* (*primary advisor role)
Under Submission, 2026
- IEEE S&P 2026 **Understanding Data Collection, Brokerage, and Spam in the Lead Marketing Ecosystem**
Yash Vekaria, Nurullah Demir, Konrad Kollnig, Zubair Shafiq
47th IEEE Symposium on Security and Privacy (*Oakland*), 2026
3rd FTC Conference on Marketing and Public Policy (*FTC*), 2026
5th CNIL Privacy Research Day (*CNIL*), 2026
Acceptance rate: 12.6% (135/1070)
- ACM CCS 2026 **Keys on Doormats: Exposed API Credentials on the Web**
Nurullah Demir, Yash Vekaria, Georgios Smaragdakis, Zakir Durumeric
33rd ACM Conference on Computer and Communications Security (*CCS*), 2026

- ACM IMC 2026 **PixelConfig: Longitudinal Measurement and Reverse-Engineering of Meta Pixel Configurations**
Abdullah Ghani, **Yash Vekaria***, Zubair Shafiq (**primary advisor role*)
26th ACM Internet Measurement Conference (IMC), 2026
- IEEE EuroS&P 2026 **Multi-Stakeholder Vulnerability Notifications in the Ad-Tech Supply Chain**
Yash Vekaria, Rishab Nithyanand, Zubair Shafiq
11th IEEE European Symposium on Security and Privacy (EuroS&P), 2026
- PETS 2026 **On the Suitability of LLM-Driven Agents for Dark Pattern Audits**
Chen Sun, **Yash Vekaria**, Zubair Shafiq, Rishab Nithyanand
26th Privacy Enhancing Technologies Symposium (PETS), 2026
Acceptance rate: 21.7% (130/706)
- ICWSM 2026 **Turning Trust to Transactions: Tracking Affiliate Marketing and FTC Compliance in YouTube's Influencer Economy**
Chen Sun, **Yash Vekaria**, Zubair Shafiq, Rishab Nithyanand
20th International AAAI Conference on Web and Social Media (ICWSM), 2026
Acceptance rate: 20%
- USENIX Security 2025 **Big Help or Big Brother? Auditing Tracking, Profiling, and Personalization in Generative AI Assistants**
Yash Vekaria, Aurelio Loris Canino, Jonathan Levitsky, Alex Ciechonski, Patricia Callejo, Anna Maria Mandalari, Zubair Shafiq
34th USENIX Security Symposium (Security), 2025
Acceptance rate: 19% (456/2400)
- Sage SM+S 2025 **Auditing the Compliance and Enforcement of Twitter's Advertising Policies**
Yash Vekaria, Zubair Shafiq, Savvas Zannettou
Social Media + Society 2025 11 (1)
- ACM IMC 2024 **Watching TV with the Second-Party: A First Look at Automatic Content Recognition in Smart TVs**
Yash Vekaria*, Gianluca Anselmi*, Alex D'Souza, Patricia Callejo, Anna Maria Mandalari, Zubair Shafiq (**joint first authors*)
24th ACM Internet Measurement Conference (IMC), 2024
Acceptance rate: 21.3% (54/253)
- IEEE S&P 2024 **The Inventory is Dark and Full of Misinformation: Understanding the Abuse of Ad Inventory Pooling in the Ad-Tech Supply Chain**
Yash Vekaria, Rishab Nithyanand, Zubair Shafiq
45th IEEE Symposium on Security & Privacy (Oakland), 2024
Acceptance rate: 17.8% (261/1463)
- ACM WebSci 2021 **Differential Tracking Across Topical Webpages of Indian News Media**
Yash Vekaria*, Vibhor Agarwal*, Pushkal Agarwal, Sangeeta Mahapatra, Sakthi Balan Muthiah, Nishanth Sastry, Nicolas Kourtellis (**joint first authors*)
13th ACM Web Science Conference, 2021
- ICWSM 2021 **Under the Spotlight: Web Tracking in Indian Partisan News Websites**
Vibhor Agarwal*, **Yash Vekaria***, Pushkal Agarwal, Sangeeta Mahapatra, Sakthi Balan Muthiah, Nishanth Sastry, Nicolas Kourtellis (**joint first authors*)
15th International AAAI Conference on Web and Social Media (ICWSM), 2021
Acceptance rate: 22.4% (30/134)
- RCIS 2021 **A Metadata-based Event Detection Method using Temporal Herding Factor and Social Synchrony on Twitter Data**
Nirmal Kumar Sivaraman, Vibhor Agarwal, **Yash Vekaria**, Sakthi Balan Muthiah
15th International Conference on Research Challenges in Information Science (RCIS), 2021

Experiences

Sep 2021 – Present **University of California, Davis**
Research Assistant, Breakerspace Lab,
Advisor: Prof. Zubair Shafiq

Davis, USA

Mar 2024 – Present	HTTP Archive <i>Open Source Contributor, Privacy, Third-party, and Generative AI Chapter</i>	Remote, USA
Summer 2024	Quantcast <i>Software Engineering Intern, Modeling Platform Infrastructure Team</i>	San Francisco, USA
Summer 2022	Max Planck Institute für Informatik <i>Visiting Research Fellow, Internet Architecture Group (INET), Advisor: Prof. Savvas Zannettou</i>	Saarbrücken, Germany
Jan 2020 – Aug 2021	Innovaccer <i>Data Scientist</i>	Noida, India

Honors & Awards

- 2025 **Distinguished Reviewer Award** *USENIX Security Symposium*
Only 23/490 PC members were awarded.
- 2024 **GGCS Ph.D. Fellowship** *University of California, Davis, USA*
Fellowship worth \$10,000 awarded by Graduate Group in Computer Science.
- 2023 **Cybersecurity Summer Fellow** *Georgia Tech, USA*
One of the 20 fellows selected from across the globe.

Teaching

- Spring 2022 **Computer Networks (ECS 152A)** *Teaching Assistant*
University of California, Davis, USA
- Spring 2019 **Data Structures & Algorithms** *Teaching Assistant*
The LNM Institute of Information Technology, India
- Fall 2018 **Computer Organization & Architecture** *Teaching Assistant*
The LNM Institute of Information Technology, India
- Fall 2017 **Computer Programming** *Teaching Assistant*
The LNM Institute of Information Technology, India

Skills

- General *Network Measurements, Computational Social Science, Data Science, Data Analytics, Data Visualization, Machine Learning, Deep Learning, Data & Text Processing, Natural Language Processing, Image Processing, Web Crawling*
- Languages *Python, Golang, Flask, HTML, Javascript, SQL/PostgreSQL, NoSQL*
- ML *Pandas, Numpy, Matplotlib, Scipy, PyTorch, OpenCV, BeautifulSoup*
- Technologies *Spark, Aerospike, Kubernetes, Docker*
- Tools *Git/GitHub, LaTeX, CI/CD, Selenium*

Leadership

- Spring 2023 **E-PRIME Mentor** *UC Davis*
Mentored undergraduate students on research project as part of the E-PRIME (Education Partnership in Research, Inclusion and Mentoring Experiences) Mentorship Program.
- Spring 2023 **MANRRS Mentor** *UC Davis*
Mentored newly-admitted undergraduate students about graduate life, career options, day-to-day academic challenges, etc. as part of the MANRRS (Minorities in Agriculture, Natural Resources, AND Related Sciences) DEI Mentorship Program.

Services

OC **Organization Committee Member**

Web Chair, ACM Internet Measurement Conference (IMC) (2026)
Web Almanac (HTTP Archive Project) (2025)

PC **Program Committee Member**

USENIX Security Symposium (2025, 2027)
Privacy Enhancing Technologies Symposium (PETS) (2026, 2027)
The Web Conference (WWW) (2026)
ACM Conference on Computer and Communications Security (CCS) (2024)
Workshop on Measurements, Attacks, and Defenses for the Web (MADWeb) (2024, 2026)
ACM Workshop on Artificial Intelligence and Security (AISec) (2023, 2024)
ACM Conference on Fairness, Accountability, and Transparency (FAccT) (2023)
ACM Web Science Conference (WebSci) (2023, 2024)
IEEE Conference on Sensing, Communication, and Networking (SECON) (2023)

AEC **Artifact Evaluation Committee Member**

Privacy Enhancing Technologies Symposium (PETS) (2024, 2025)
Network and Distributed System Security Symposium (NDSS) (2024)
IEEE Workshop on Offensive Technologies (WOOT) (2023)

Invited Talks

- Mar 2026 **Memory and Personalization in LLMs**
Rising Stars in CS Colloquium, University of Iowa, Iowa City, IA
Center for Cybersecurity Seminar, New York University Abu Dhabi, Virtual
- Mar 2026 **Understanding Data Collection, Brokerage, and Spam in the Lead Marketing Ecosystem**
Federal Trade Commission (FTC), Washington D.C.
- Feb 2026 **Memory and Personalization in LLMs**
CITRIS Tech Policy Workshop, UC Berkeley, CA
- Sep 2025 **Understanding Opaque Data Practices and Risks in Online Marketing and User Personalization**
Stanford Security Lunch, Stanford University, Palo Alto, CA
- Jul 2024 **Uncovering hidden security and privacy threats in online advertising**
University College London (UCL), United Kingdom
- May 2023 **Be Aware: Dark Pooling Uncovered**
AdExchanger's Programmatic I/O, MGM Las Vegas
- Jul 2021 **Differential Tracking Across Topical Webpages of Indian News Media**
CMLBDA, The LNM Institute of Information Technology, India

Selected Media Coverage

- May 2026 **Chatbot Ad IDs Share Data To Google, Microsoft, Other Analytics Providers**
[MediaPost](#) article by *Laurie Sullivan*
- Mar 2026 **Security boffins scoured the web and found hundreds of valid API keys**
[The Register](#) article by *Thomas Claburn*
- Mar 2026 **Security credentials inadvertently leaked on thousands of websites**
[New Scientist](#) article by *Chris Stokel-Walker*
- Oct 2025 **AI Browsers are a Cybersecurity Time Bomb**
[The Verge](#) article by *Robert Hart*
- June 2025 **Yes, Your TV Is Probably Spying on You. Your Fridge, Too. Here's What They Know.**
[NY Times](#) article by *Rachel Cericola, Jon Chase and Lee Neikirk*

- Mar 2025 **Roku, Amazon, and practically every company in the streaming business are inventing new ways to make money off your data.**
[Vox](#) article by *Adam Clark Estes*
You know that generative AI browser assistant extension is probably beaming everything to the cloud, right?
[The Register](#) article by *Thomas Claburn*
- Dec 2024 **Yes, Your TV Is Probably Spying on You. Your Fridge, Too. Here's What They Know.**
[Medium](#) article by *Kamal Acharya*
- Nov 2024 **You think you're watching smart TV ... In fact it's watching you**
[The Times](#) article by *Mark Sellman*
- Sep 2024 **Smart TVs take snapshots of what you watch multiple times per second**
[New Scientist](#) article by *Jeremy Hsu*
- Sep 2023 **Elon Musk's X is ignoring its own ad policies, a new study finds**
[Fast Company](#) article by *Chris Stokel-Walker*